

Problem 1

If a and b are complex numbers, prove that:

- 1° $|ab| = |a||b|, \quad |a/b| = |a|/|b| \quad (b \neq 0);$
- 2° $\operatorname{Re}(a) = \frac{1}{2}(a + \bar{a}), \quad \operatorname{Im}(a) = \frac{1}{2i}(a - \bar{a});$
- 3° $\overline{a \pm b} = \bar{a} \pm \bar{b}, \quad \overline{ab} = \bar{a}\bar{b}, \quad \overline{(a/b)} = \bar{a}/\bar{b};$
- 4° $\operatorname{Arg}(ab) = \operatorname{Arg} a + \operatorname{Arg} b, \quad \operatorname{Arg}(a/b) = \operatorname{Arg} a - \operatorname{Arg} b;$
- 5° $||a| - |b|| \leq |a + b| \leq |a| + |b|.$

Solution

Part 1°: Modulus of Product and Quotient

Prove: $|ab| = |a||b|$

Let $a = x_1 + iy_1$ and $b = x_2 + iy_2$.

Method 1 (Direct calculation):

$$ab = (x_1 + iy_1)(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)$$

$$|ab|^2 = (x_1x_2 - y_1y_2)^2 + (x_1y_2 + x_2y_1)^2$$

Expanding:

$$= x_1^2x_2^2 - 2x_1x_2y_1y_2 + y_1^2y_2^2 + x_1^2y_2^2 + 2x_1x_2y_1y_2 + x_2^2y_1^2$$

$$= x_1^2x_2^2 + y_1^2y_2^2 + x_1^2y_2^2 + x_2^2y_1^2$$

$$= x_1^2(x_2^2 + y_2^2) + y_1^2(x_2^2 + y_2^2) = (x_1^2 + y_1^2)(x_2^2 + y_2^2)$$

$$= |a|^2|b|^2$$

Taking square roots: $|ab| = |a||b|$

Method 2 (Using conjugates):

$$|ab|^2 = (ab)\overline{(ab)} = ab\bar{a}\bar{b} = a\bar{a} \cdot b\bar{b} = |a|^2|b|^2$$

Therefore $|ab| = |a||b|$

Prove: $|a/b| = |a|/|b| \quad (b \neq 0)$

From $|ab| = |a||b|$, let $a' = a/b$. Then:

$$|a| = |a' \cdot b| = |a'||b|$$

Therefore:

$$|a'| = \frac{|a|}{|b|} \implies \left| \frac{a}{b} \right| = \frac{|a|}{|b|}$$

Part 2°: Real and Imaginary Parts

Prove: $\operatorname{Re}(a) = \frac{1}{2}(a + \bar{a})$

Let $a = x + iy$. Then $\bar{a} = x - iy$.

$$a + \bar{a} = (x + iy) + (x - iy) = 2x$$

$$\frac{1}{2}(a + \bar{a}) = x = \operatorname{Re}(a)$$

Prove: $\text{Im}(a) = \frac{1}{2i}(a - \bar{a})$

$$a - \bar{a} = (x + iy) - (x - iy) = 2iy$$

$$\frac{1}{2i}(a - \bar{a}) = \frac{2iy}{2i} = y = \text{Im}(a)$$

Part 3°: Conjugate Properties

Prove: $\overline{a \pm b} = \bar{a} \pm \bar{b}$

Let $a = x_1 + iy_1$ and $b = x_2 + iy_2$.

$$a \pm b = (x_1 \pm x_2) + i(y_1 \pm y_2)$$

$$\overline{a \pm b} = (x_1 \pm x_2) - i(y_1 \pm y_2)$$

On the other hand:

$$\bar{a} \pm \bar{b} = (x_1 - iy_1) \pm (x_2 - iy_2) = (x_1 \pm x_2) - i(y_1 \pm y_2)$$

Therefore $\overline{a \pm b} = \bar{a} \pm \bar{b}$

Prove: $\overline{ab} = \bar{a} \bar{b}$

$$ab = (x_1 + iy_1)(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)$$

$$\overline{ab} = (x_1x_2 - y_1y_2) - i(x_1y_2 + x_2y_1)$$

Now:

$$\bar{a} \bar{b} = (x_1 - iy_1)(x_2 - iy_2) = (x_1x_2 - y_1y_2) - i(x_1y_2 + x_2y_1)$$

Therefore $\overline{ab} = \bar{a} \bar{b}$

Prove: $\overline{(a/b)} = \bar{a}/\bar{b}$

From $\overline{ab} = \bar{a} \bar{b}$, let $c = a/b$. Then $a = bc$.

$$\bar{a} = \overline{bc} = \bar{b} \bar{c}$$

Therefore:

$$\bar{c} = \frac{\bar{a}}{\bar{b}} \implies \overline{(a/b)} = \frac{\bar{a}}{\bar{b}}$$

Part 4°: Argument Properties

Prove: $\text{Arg}(ab) = \text{Arg } a + \text{Arg } b$

Write in polar form:

$$a = r_1(\cos \theta_1 + i \sin \theta_1) = r_1 e^{i\theta_1}$$

$$b = r_2(\cos \theta_2 + i \sin \theta_2) = r_2 e^{i\theta_2}$$

Then:

$$ab = r_1 r_2 e^{i(\theta_1 + \theta_2)} = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

Therefore:

$$\text{Arg}(ab) = \theta_1 + \theta_2 = \text{Arg } a + \text{Arg } b$$

(modulo 2π)

Prove: $\text{Arg}(a/b) = \text{Arg } a - \text{Arg } b$

$$\frac{a}{b} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

Therefore:

$$\text{Arg}(a/b) = \theta_1 - \theta_2 = \text{Arg } a - \text{Arg } b$$

(modulo 2π)

Part 5°: Triangle Inequalities

Prove: $|a + b| \leq |a| + |b|$ (**Triangle Inequality**)

$$\begin{aligned} |a + b|^2 &= (a + b)\overline{(a + b)} = (a + b)(\bar{a} + \bar{b}) \\ &= a\bar{a} + a\bar{b} + b\bar{a} + b\bar{b} = |a|^2 + |b|^2 + a\bar{b} + \overline{a\bar{b}} \\ &= |a|^2 + |b|^2 + 2\text{Re}(a\bar{b}) \end{aligned}$$

Since $\text{Re}(z) \leq |z|$:

$$|a + b|^2 \leq |a|^2 + |b|^2 + 2|a\bar{b}| = |a|^2 + |b|^2 + 2|a||b| = (|a| + |b|)^2$$

Therefore $|a + b| \leq |a| + |b|$

Prove: $||a| - |b|| \leq |a + b|$ (**Reverse Triangle Inequality**)

From $|a| = |(a + b) - b| \leq |a + b| + |b|$:

$$|a| - |b| \leq |a + b|$$

Similarly, from $|b| = |(a + b) - a| \leq |a + b| + |a|$:

$$|b| - |a| \leq |a + b|$$

Combining these:

$$||a| - |b|| \leq |a + b|$$

Final Answer

All five properties have been proven:

- 1° $|ab| = |a||b|$, $|a/b| = |a|/|b|$
- 2° $\text{Re}(a) = \frac{1}{2}(a + \bar{a})$, $\text{Im}(a) = \frac{1}{2i}(a - \bar{a})$
- 3° $\overline{a \pm b} = \bar{a} \pm \bar{b}$, $\overline{a\bar{b}} = \bar{a}b$, $\overline{(a/b)} = \bar{a}/\bar{b}$
- 4° $\text{Arg}(ab) = \text{Arg } a + \text{Arg } b$, $\text{Arg}(a/b) = \text{Arg } a - \text{Arg } b$
- 5° $||a| - |b|| \leq |a + b| \leq |a| + |b|$